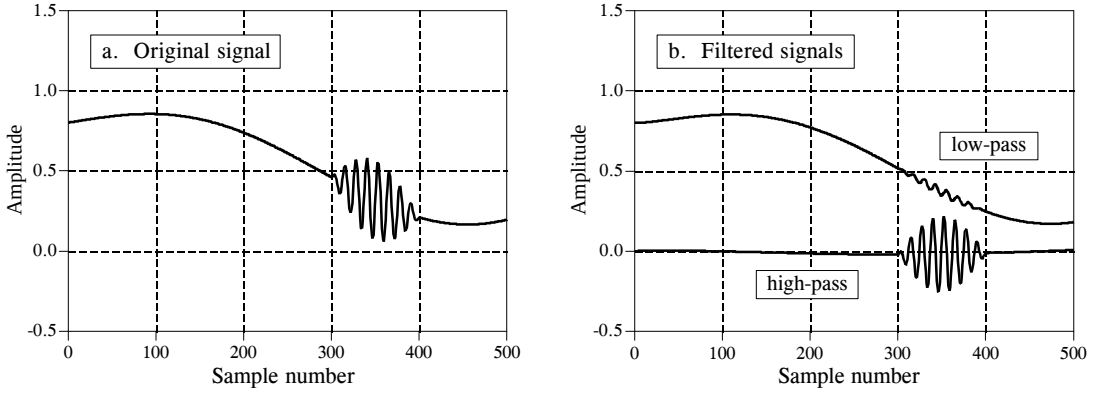

FIGURE 19-4
Example of single pole recursive filters. In (a), a high frequency burst rides on a slowly varying signal. In (b), single pole low-pass and high-pass filters are used to separate the two components. The low-pass filter uses $x = 0.95$, while the high-pass filter is for $x = 0.86$.

filter becomes *unstable* if x is made greater than one. That is, any nonzero value on the input will make the output increase until an overflow occurs.

The value for x can be directly specified, or found from the desired *time constant* of the filter. Just as $R \times C$ is the number of seconds it takes an RC circuit to decay to 36.8% of its final value, d is the number of samples it takes for a recursive filter to decay to this same level:

EQUATION 19-4
Time constant of single pole filters. This equation relates the amount of decay between samples, x, with the filter's time constant, d, the number of samples for the filter to decay to 36.8%.

$$x = e^{-1/d}$$

For instance, a sample-to-sample decay of $x = 0.86$ corresponds to a time constant of $d = 6.63$ samples (as shown in Fig 19-3). There is also a fixed relationship between x and the -3dB *cutoff frequency*, f_C, of the digital filter:

EQUATION 19-5
Cutoff frequency of single pole filters. The amount of decay between samples, x, is related to the cutoff frequency of the filter, f_C, a value between 0 and 0.5.

$$x = e^{-2\pi f_C}$$

This provides three ways to find the "a" and "b" coefficients, starting with the time constant, the cutoff frequency, or just directly picking x.

Figure 19-4 shows an example of using single pole recursive filters. In (a), the original signal is a smooth curve, except a burst of a high frequency sine wave. Figure (b) shows the signal after passing through low-pass and high-pass filters. The signals have been separated fairly well, but not perfectly, just as if simple RC circuits were used on an analog signal.